SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL3–TechnicalPresentation

CourseCode:	CSA0305	**CourseTitle:**	DataStructures
COAssessed:	CO1–Imitation(BL3,BL4)	**Assessment:**	AssessmentTool3–Technical Presentation
Weightage:	35%ofCIA	**TotalMarks:**	100
CO1Statement:	*ApplysuitabledatastructuresandADTsforefficientproblemsolvingbyanalyzingtimeandspace complexity.*		
Student Name:	M.VAISHNAV	**Reg. No.:**	192524251
Section/Batch:		**Date:**	15/07/2026

CodeofConduct

I M.VAISHNAV (192524251) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature ofthe Student: *M.VAISHNAV (192524251)*

Instructions

- Thisassessmentcontainstechnicalpresentation.Total:100Marks.
- Whena student delivers a technical presentation on a data structures topic,evaluation should be based onthefollowingrequirements:Conceptexplanation,Real-worldapplicationandClarityofpresentation.
- Demonstrateclearunderstanding,notjustreadingslides
- Useoftools:PowerPoint/Google Slides.
- Duration:7–10minutespresentation+3minutesQ&A.

Section/Topic	Focus	QNos	Marks
DataStructure	Real-TimeApplications	1	100
	GRANDTOTAL:	**100Marks**	

AnnexureA–TechnicalPresentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordereddynamically.Recommendasuitabledatastructureandjustifybasedonflexibilityand efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station.Suggestasuitabledatastructureforrepresentationandjustifyyouranswer.(100Mark)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency.(100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

RubricsforCalculation

Criteria	Weightage	Level4–Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
ConceptUnderstanding	20	Demonstratesdeepandaccurateunderstandingof data structure withinsightsandconnections	Showsclearunderstandingwithminorgaps	Basicunderstanding; somemisconceptions present	Limitedorincorrectunderstanding
ContentOrganization	30	Logical,well-structured,smooth flow;strongintroductionandconclusion	Mostlyorganized with clear flow	Someorganization butlacksclarityortransitions	Disorganized,difficulttofollow
TechnicalDepth	20	Includesalgorithms,complexityanalysis,andadvancedinsights	Includesalgorithmand basiccomplexityanalysis	Limitedtechnical details;missingdepth	Minimal or notechnicalcontent
PowerPointSlides	20	Excellentdesign,clearvisuals,enhancesunderstanding	Goodslideswithminorissues	Basicslides;somewhatclutteredorunclear	Poorslidesornovisualsupport
CommunicationSkills	10	Clear,confident,engagingdelivery;excellentarticulation	Cleardeliverywith minorhesitation	Somehesitation; unclear in parts	Poorcommunication;difficulttounderstand

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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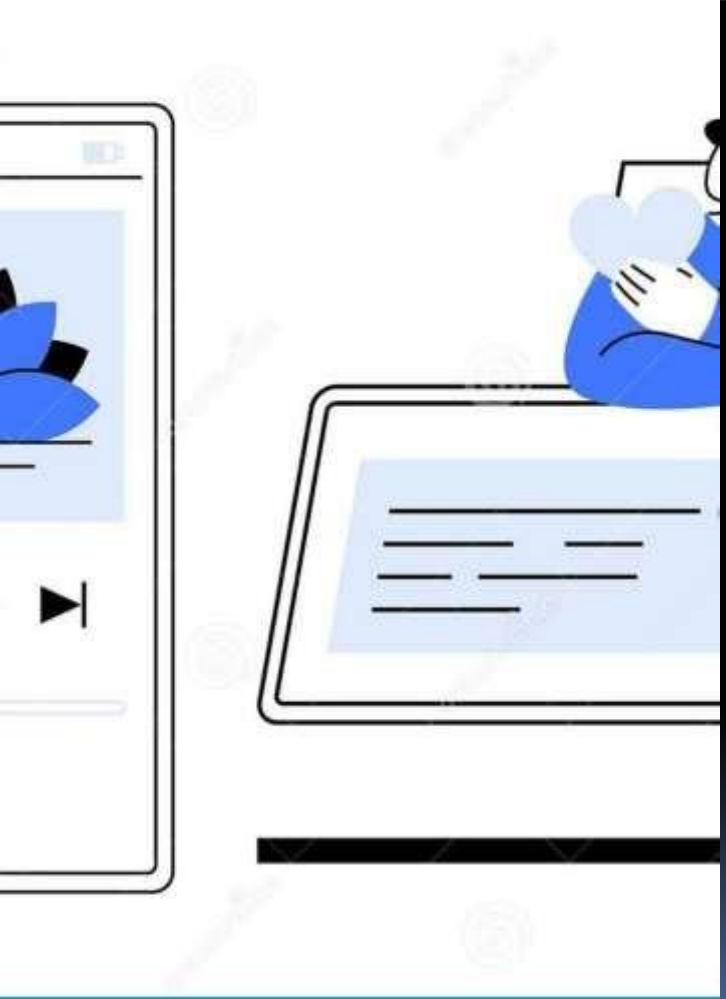
EfficientPlaylistOperationsUsingDoublyLinkedLists

Presentedby:M.VAISHNAV—RollNo:192524251—Dept.ofArtificial IntelligenceandDataScience,SIMATSEngineering



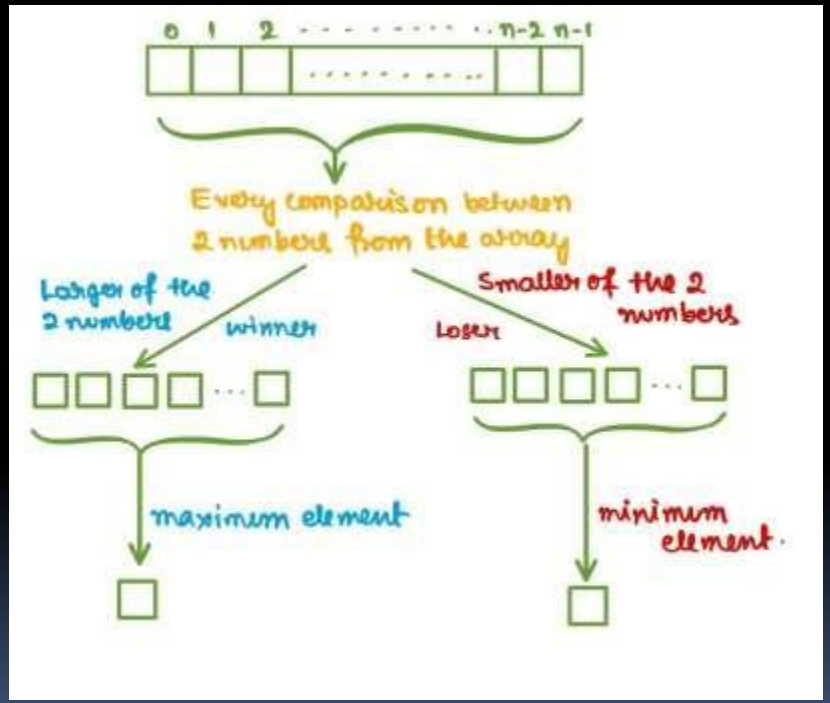
Intío®ustion — WhK plΘKlist peífoímΘnse mΘtteís

Streamingservicesmanagelargelibrariesandmillionsofuserplaylists. Usersexpectinstantresponseswhentheyadd,remove,orreordertracks. Theplaylistback-endmustprovidelow-latencyoperationswhile supportingfeatureslikeskip,repeat,andhistory.

PíoblemstΘtement

- Supportdynamicinsertionoftracksatarbitrarypositions
- Allowdeletionwithminimaloverhead
- Enablereorderingandmoveswithconstant-timeupdates
- ProvidefastNext/Previousnavigationforplayback
- Scaleefficientlyforverylongplaylists

ÐΘtΘStíustuíeCΘn®i®Θtes



6ííΘK

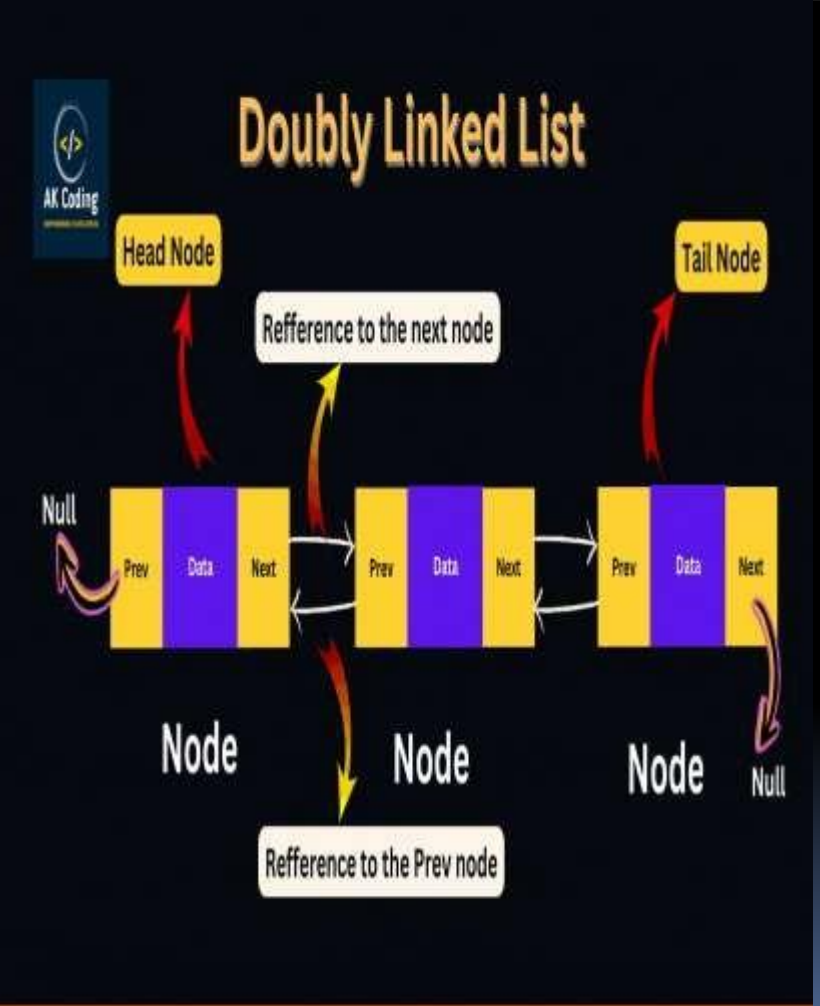
Pros:O(1)randomaccess,compactmemorylayout.Cons: insertion/deletionrequireshiftingelements—O(n)worst-case; expensivefor frequent edits.

Sin⑦lKLinke®List

Pros:O(1)insertionatknownnode,low-costnodeallocation. Cons:nobackwardtraversal,deletionrequirespointertoprevious node or O(n) search.

ÐoublKLinke®List

Pros:O(1)insertion&deletiongivennodereference,forwardand backwardtraversal,idealforplayerNext/Previousandefficient reordering.Slightlyhighermemory(extrapointer)butbetter operationalfit for playlists.



Resommen®e®Stíustuíe—ÐoubIKLinke®List

Eachplaylistnodestores:previouspointer,songmetadata(id,title,artist,duration,URI),andnextpointer.Thelistsupportshead/tailreferencesandoptionalhashing(indexmap)forO(1)lookupofnodesbysongid.

No®e sontents

Prev pointer · Songdata·Next pointer

6uxiliΘíK stíustuíes

Hashmap:songId →node(optional) fordirectaccess

PlΘKlist feΘtuíes

Skip,repeat, move-to-front, remove,insert-after,insert-before

WhKÐoublKLinke®ListsWin

Bi-®iíestionΘln Θvi⑦Θtion

NativesupportforNext andPreviousplayback without extra state.

ConstΘnt-time e®its

Insert,remove,andmove require only pointer updates when node referenceknown.

Inte⑦íΘteswith sKns&sΘshin⑦

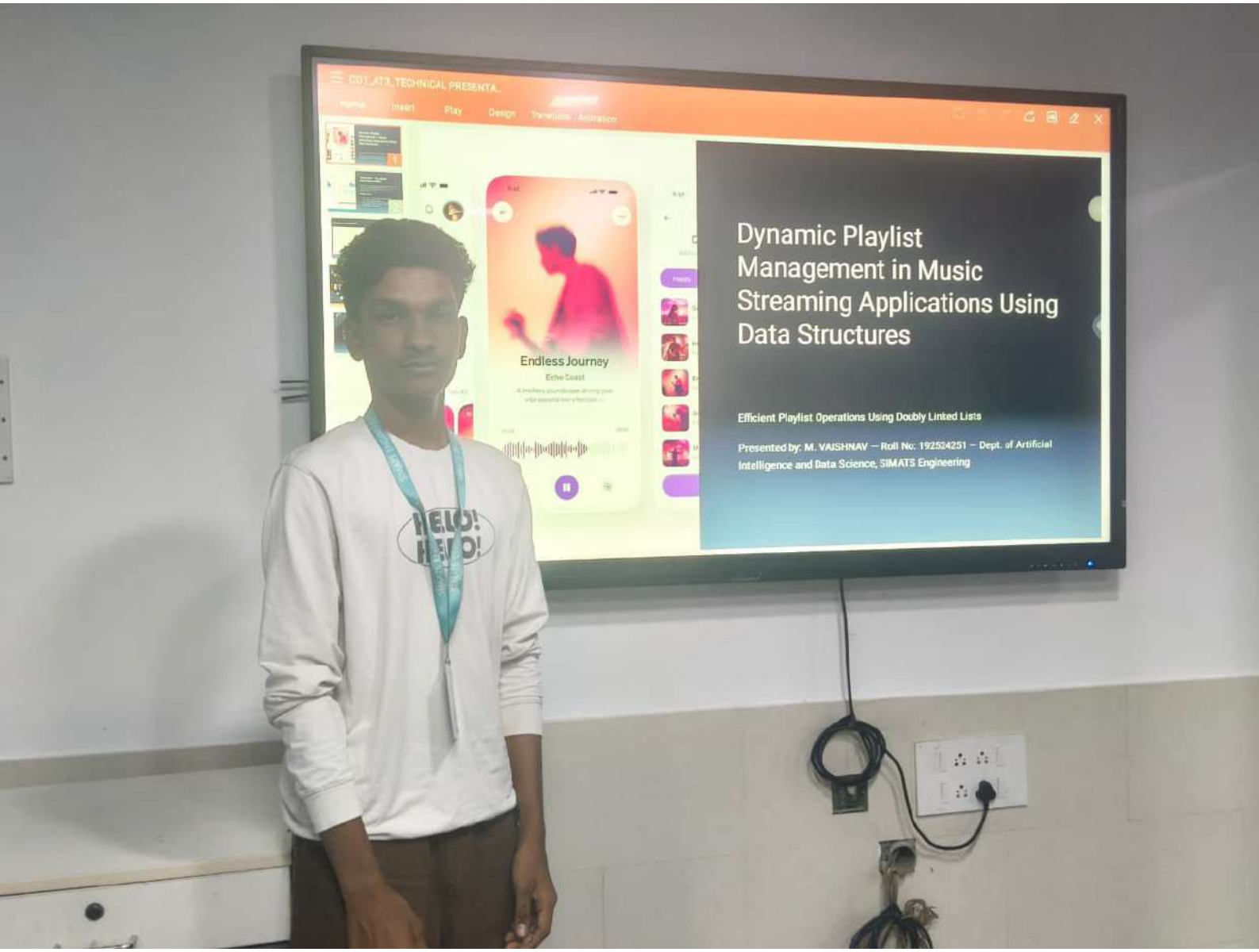
Works well with diffs, CRDTs, and server-side persistenceforlarge-scale apps.

Conslusion &Resommen® Θtion

Doublylinkedlistsprovidethebestbalanceofefficiency,flexibility,and straightforwardimplementationfordynamicplaylistmanagementinmusic streamingapps.Whencombinedwithanoptionalsong Id→nodemapanda compactsynclog,theydeliverO(1)runtimeeditsandrobustcross-device behavior.

Finalrecommendation: Implementanin-memorydoublylinkedlistfor runtimeplaylistoperations,augmentwithanindexmapfordirectaccess, andintegratealightweighteventlogforsynchronizationanddurability.

Thankyou


CD1_AT3_TECHNICAL PRESENTA...
Endless Journey
Dynamic Playlist Management in Music Streaming Applications Using Data Structures
Efficient Playlist Operations Using Doubly Linked Lists
Presented by: M. VAISHNAV — Roll No: 192524251 — Dept. of Artificial Intelligence and Data Science, SIMATS Engineering